

Polluted Black Holes using 3-D Weighted Monte Carlo Models

Raghav Chari^{(1),*} Apurva Oza^{(2),†} and Moritz Meyer zu Westram^{(3),‡}

⁽¹⁾New York University, New York, NY 10012, USA

⁽²⁾Jet Propulsion Laboratory,

California Institute of Technology, Pasadena, CA 91109, USA

⁽³⁾Physikalisches Institut, Universität Bern, Bern, Switzerland

(Dated: April 29, 2026)

Mass accretion is generally modeled as a two-body system with a satellite M_2 accreting $\dot{M}$ onto a primary M_1 , the central object. Studying the physical mechanisms generically across a range of masses, provides insight into the energetics of such systems in the Universe. We consider high-mass objects orbiting a primary (e.g., star-black hole) as well as low-mass objects orbiting primaries (e.g., exomoon-exoplanet) focusing on the metallicity χ which can in principle, be observed. Test cases include the presence of metals in gas giants, as well as the accretion disks around black holes. Our research explores the concept of tidal disruption events (TDEs) and the associated accretion disk formation for all objects. The study combines numerical simulations and analytic models, with the aim of making robust predictions on the particles in gas giant atmospheres, products which may be detectable spectrally today. Our results may inform future observations and detection strategies, and deepen our understanding of TDEs, exomoon composition, and the complex dynamics of accretion processes. At the larger scale, we find that the magnetospheric physics governing exoplanet-exomoon interactions may be informed by black holes as well.

PACS numbers:

I. INTRODUCTION

Our nearest black hole is Gaia BH1, a $\sim 10M_\odot$ dormant black hole in a binary with a Sun-like companion at a distance of only 480 pc [1]. . AO (26 Oct 2025): *Read paper and study metallic profile as benchmark - and examine binaries, and $\gtrsim$ - 2 body systems in general. Need spectra*

The proximity of Gaia BH1 motivates a quantitative framework for predicting the chemical and dynamical signatures expected from material orbiting such a compact object. Unlike polluted white dwarfs, accreting black holes do not retain absorbers in a photosphere; instead, the relevant observables are encoded in the spatial and velocity distribution of orbiting material, set by general-relativistic dynamics down to the innermost stable circular orbit (ISCO).

In one paragraph: SERPENS is a 3D weighted-Monte-Carlo test-particle code originally developed for trace material lost from exo-Ios into exoplanet atmospheres [2, 3]. Each superparticle carries a species, a launch velocity drawn from a physically-motivated distribution (thermal evaporation, sputtering), and a weight set by the source mass-loss rate. We extend this framework to compact-object accretion by (i) including 1PN gravity through REBOUNDx [4, 5] so dynamics remain valid down to a few r_s , (ii) introducing an absorbing inner boundary at the ISCO [6], and (iii) adding a stellar-wind launch mode driven by Parker-type isothermal coronal expansion [7, 8] so that a mass-losing companion can serve as the source of trace material. The output is a phase-resolved spatial-and-velocity distribution from which we extract analytical Bondi–Hoyle–Lyttleton capture diagnos-

tics [9], sky-projected column densities at the binary inclination, and Doppler line profiles. Gaia BH1 is our proof-of-concept system; the framework is general and can be applied to any star–BH binary or to material near isolated compact accretors [10].

The phenomenon of pollution in stellar bodies, especially in white dwarfs, has garnered significant scientific interest in recent years. Studies like those conducted by Doyle et al. (2021) [11] have played a pivotal role in deepening our understanding of this process. Stellar pollution typically involves the absorption of celestial bodies, such as planets and moons, into stars. This absorption significantly impacts the stellar composition and can provide valuable insights into the history and evolution of these celestial systems.

An intriguing aspect of this phenomenon is observed in the case of exo-Ios. These are moons with toroidal atmospheres, often around exoplanets. Oza et al. (2019) [3] demonstrates that these exo-Ios, which exhibit volcanic activity, contribute to the pollution of their stellar environments. This finding underscores the complexity of interactions within stellar systems and the varied sources of stellar pollution.

While much of the research on stellar pollution has focused on stars, black holes present a markedly different scenario. Their strong gravitational pulls and the presence of event horizons introduce unique dynamics compared to typical stars. This distinction leads to intriguing questions regarding the fate of celestial bodies, particularly exomoons, when they are captured by black holes.

II. POLLUTED BLACK HOLES

The study of accretion disks around black holes is essential to understanding the complex physics of matter accumulation in these massive celestial bodies (e.g., Pringle et al. [12], Blandford and Payne [13], Artimage[14]). Foundational work

*Electronic address: rc6116@nyu.edu

†Electronic address: apurva.oza.space@gmail.com

‡Electronic address: moritz.meyerzuwestram@unibe.ch

in this area, (eg., Armitage [14], Rees, [15]) and others, illuminates the interplay of gravitational, centrifugal, and viscous forces within these disks. This interplay results in differential rotation and the inward spiraling of matter toward the central black hole.

Near black holes, the dynamics of accretion disks are further complicated by the intense gravitational fields and the consequential relativistic effects. These factors make the study of matter accumulation near black holes, particularly the accretion of exomoons, a complex and fascinating subject.

Kinetic-regime applicability. Because SERPENS is a collisionless test-particle code, we briefly note when this treatment is appropriate near a black hole. The diagnostic is the Knudsen number $\text{Kn} = \lambda_{\text{mfp}}/L$: fluid magnetohydrodynamics is justified for $\text{Kn} \ll 1$ (e.g. the optically-thick disk midplane of [16]), whereas a kinetic / test-particle treatment is required when $\text{Kn} \gtrsim 1$ or when species are sub-dominant. The kinetic regime is well-established for hot low-density accretion flows [17, 18], for trace ions in stellar winds [19], and for Bondi–Hoyle–Lyttleton capture in low-density environments [9, 10]. For Gaia BH1 specifically, the G-dwarf coronal launch is collisional ($n \sim 10^9 \text{ cm}^{-3}$), but the wind density falls as r^{-2} so that across most of the binary volume the bulk wind is collisionless against itself; trace metals are sub-dominant throughout. We therefore adopt a test-particle treatment for the wind material and explicitly defer the optically-thick inner accretion disk to standard MHD.

Two relativistic length scales bound the dynamically relevant region around a non-spinning black hole. The Schwarzschild radius $r_s = 2GM/c^2$ defines the event horizon, while the innermost stable circular orbit (ISCO) lies at $r_{\text{ISCO}} = 3r_s$ [6]. Test particles on bound orbits with periaapsis below the ISCO cannot maintain stable circular motion and plunge across the horizon on a dynamical timescale. For our reference $10M_\odot$ stellar-mass black hole, $r_s \approx 29.5 \text{ km}$ and $r_{\text{ISCO}} \approx 88.6 \text{ km}$. Outside the ISCO, the dominant relativistic correction to Newtonian dynamics is periastron precession, which scales as $\Delta\varpi \propto (r_s/a)$ per orbit at first post-Newtonian (1PN) order [20]. Capturing both effects in a Monte Carlo particle simulation requires modifying the standard N -body integration to include 1PN corrections and an absorbing inner boundary at the ISCO.

Understanding the dynamics of accretion disks around black holes is crucial for analyzing how exomoons are captured and potentially accreted by these massive celestial bodies. This study aims to explore the potential evidence of icy exomoons falling into black hole accretion disks and their consequent pollution. By examining this process, we can gain insights into the dynamic environment surrounding black holes and possibly identify observational signatures indicative of exomoons in these extreme environments. This novel approach extends the concept of spallogenic nuclides, previously associated with the accretion of icy exomoons in white dwarfs, to the context of black holes.

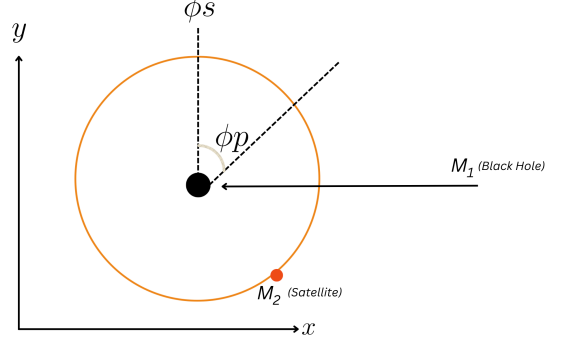


FIG. 1: SERPENS utilizes a distinct geometrical and coordinate framework. Initially, the x-axis coincides with the alignment of the exoplanet and black hole at $t = 0$. As time proceeds, the exoplanet traverses its orbital path, governed by the true anomaly ϕ_p . The y-axis, situated within the orbital plane, stands orthogonal to the x-axis. Concurrently, the z-axis is aligned parallel to the orbital plane’s specific angular momentum vector. The spatial position of the exomoon mirrors that of the exoplanet’s coordinate system, albeit adjusted for the exoplanet’s location. To effectively chart the exomoon’s path, a reference frame that rotates in unison with the exoplanet is utilized.

III. MATERIALS AND METHODS

A. SERPENS

Meyer zu Westram et al. 2023 (submitted) [2] studies the random velocity distribution of particles in free space. SERPENS employs a test particle algorithm on a 4-D grid (r, ϕ, z, t).

1. Planetary Accretion

We can approximate accretion as energy-limited Escape from an Exo-Io (Watson et al.[21], demonstrated by atmospheric escape in the Kepler data (Jin et al. [22] and Fulton & Petigura[23]. This results in the hydrodynamic escape approximated by Oza et al. [3]

$$\dot{M}_U \sim m_i x_i \frac{Q}{U_s (R_a)}, \quad (1)$$

where,

$$Q = \eta_{\text{XUV}} 4\pi R_a^2 F_{\text{XUV}}, \quad (2)$$

Note that we can use $\eta_{\text{XUV}} = 0.1 - 0.4$ (eg., Murray-Clay et al. [24]). The escape flux may also be reduced if Q_{net} is too large as demonstrated by (Johnson et al. [25]):

$$Q_{\text{net}} > Q_c \approx 4\pi r_* \frac{\gamma}{c_c \sigma_c K n_m} \sqrt{\frac{2U(r_*)}{m}} U(r_0), \quad (3)$$

2. Employing Particles

For our analysis we employ a suite of particles formulated by Meyer zu Westram et al 2023 (submitted) [2]. The velocity distribution follows a maxwell boltzman distribution with scaled parameter v_{sc} given by the distribution of χ^2 .

$$f_{\text{th}}(v; \mathbf{r}) = \sqrt{\frac{2}{\pi}} \frac{v^2}{v_{\text{sc}}^3} \exp\left(-\frac{v^2}{v_{\text{sc}}^2}\right), \quad (4)$$

where,

$$v_{\text{sc}}(\mathbf{r}) = \sqrt{\frac{k_B T}{m}}, \quad (5)$$

The mass loss is determined by the ejection of particles p. One such model was represented by Smyth and Combi [26] that combines the Boltzman distribution with a high-energy tail:

$$f_{\text{sp}}(v; \alpha, v_b, v_M) = \frac{1}{D(\alpha, \frac{v_M}{v_b})} \frac{1}{v_b} \cdot \left(\frac{v}{v_b}\right)^3 \left(\frac{v_b^2}{v^2 - v_b^2}\right)^\alpha \cdot \left(1 - \sqrt{\frac{v^2 + v_b^2}{v_M^2}}\right), \quad (6)$$

where we have the dimensionless normalization constant

$$D(\alpha, v_M/v_b) = \int_0^{\sqrt{(v_M/v_b)^2 - 1}} \frac{x^3}{(1+x^2)^\alpha} \left(1 - \frac{v_b}{v_M} \sqrt{1+x^2}\right) dx, \quad (7)$$

3. Mass Loss

SERPENS simulates reactions defined by their reaction rate at each advance, and the weight of a superparticle calculated as follows:

$$w(t) = e^{-t/\tau_1} e^{-t/\tau_2} \dots e^{-t/\tau_n}, \quad (8)$$

with given lifetime $\tau_i = 1/v_i$ for the i -th reaction given the reaction rate v_i [2]

These integrations are conducted through the REBOUND software (Rein and Liu [27]). Particles that are collided with an object due to gravitational interaction, would be therefore removed from analysis.

4. Calculating Mixing Ratios

The settling speed can be used to find the mixing ratio through the following relation from (Arras et al. [28]):

$$\frac{n}{n_b} = \frac{F}{u n_b} = -\frac{F v}{g n_b} = -\frac{F \langle \sigma v \rangle}{g} \quad (9)$$

where $F < 0$ for downward motion. Since the background density has canceled out of the right-hand side, the result will vary slowly with altitude.

The particle flux below the source can be written as:

$$F = -f_{\text{pl}} \epsilon_{\text{abl}} f \frac{\dot{M}_d}{4\pi R_p^2 \bar{m}} \simeq -6.5 \times 10^9 \text{ cm}^{-2} \text{ s}^{-1} f \left(\frac{f_{\text{pl}} \epsilon_{\text{abl}} \dot{M}_d}{10^8 \text{ g s}^{-1}} \right) \left(\frac{R_p}{R_J} \right)^{-2}. \quad (10)$$

Here, $\dot{M}_d$ is the accretion rate of the dust population as it moves toward the star under PR drag, f is the abundance by number of this element in the dust particles, and $\bar{m}$ is the mean mass of the elements making up the dust particles.

The results are now assembled to compute the mixing ratio below the meteoric source. The result is:

$$\frac{n}{n_b} = 4.1 \times 10^{-3} f \left(\frac{f_{\text{pl}} \epsilon_{\text{abl}} \dot{M}_d}{10^8 \text{ g s}^{-1}} \right) \times \left(\frac{M_p}{M_J} \right)^{-1} \left(\frac{T}{10^3 \text{ K}} \right)^{1/2} \left(\frac{m_b m_u}{m(m + m_b)} \right)^{1/2}. \quad (11)$$

The relative number of different elements scales as f/m for $m > m_b$, and favor small planets due to the smaller gravity.

5. Estimating the number density of the background gas

The background gas n_b can be written as if the gas is ideal:

$$n_b \sim P_{b,\text{atm}} / k_b / T_{b,\text{atm}} \quad (12)$$

Where we assume that $T_{b,\text{atm}} \sim T_{\text{eq}}$ of the exoplanet atmosphere, and the maximum gas pressure probed in transmission is roughly $P_{b,\text{atm}} \sim 1 \text{ mbar}$.

B. General Relativistic Corrections

To extend SERPENS from the planetary regime to the strong-field environment of a stellar-mass black hole, we incorporate post-Newtonian gravitational corrections through the `gr_full` force module of REBOUNDx [4], which implements the 1PN equations of motion of Newhall, Standish & Williams [5]. These equations include all interbody pairwise terms scaling as $(v/c)^2$ and (GM/rc^2) , and reproduce the precession of Mercury and the Hulse–Taylor binary at the part-per-million level [20]. Because the test particles in our simulation are non-self-gravitating and gravitationally subdominant,

only the central black hole is designated as a relativistic source via the `gr_source` flag.

C. Stellar Wind Launch and Photoionization Lifetimes

For star–BH binaries the natural source of trace material is the stellar companion. We add a wind launching mode that draws each particle’s velocity from a shifted Maxwellian: a Maxwell–Boltzmann thermal scatter at the coronal temperature T_{corona} , plus a deterministic radial bulk velocity v_{∞} . This is the standard kinetic representation of a thermally-driven Parker wind sampled at the launch surface ([7]; [8] Ch. 5; observationally validated in the in-situ solar wind by [19]). For a solar-analog G-dwarf we adopt $T_{\text{corona}} = 1.5 \times 10^6$ K and derive $v_{\infty} \approx 4c_s \approx 400$ km s $^{-1}$ from the isothermal sound speed. Total mass-loss rates are calibrated against the Wood et al. [29] astrospheric Ly α measurements, scaled by stellar surface area and a Skumanich-like activity factor; for our reference G-dwarf this yields $\dot{M} \approx 10^9$ kg s $^{-1}$ ($\approx 2 \times 10^{-14} M_{\odot}$ yr $^{-1}$). A more complete treatment using Alfvén-wave-driven wind models [30] is straightforward to substitute.

Trace species are lost from the simulation according to a per-species photoionization lifetime $\tau = (\sigma_X \Phi_X)^{-1}$, where Φ_X is the ionizing photon flux from the BH accretion disk and σ_X is an effective species cross section taken from [31]. For Gaia BH1’s observational X-ray upper limit ($L_X \lesssim 10^{29}$ erg s $^{-1}$, [1]) the resulting lifetimes range from $\sim 10^{11}$ s for H to $\sim 10^8$ s for highly-ionized metals at the binary separation, comparable to or longer than the orbital period for hydrogen and shorter for inner-shell ions. The framework therefore naturally produces species-dependent observational signatures even in the dormant regime.

Particle removal proceeds through two boundaries. The outer boundary, at $r_{\text{max}} a_{\text{src}}$ from the orbital primary (with a_{src} the source body’s semi-major axis), removes particles that are dynamically unbound from the system. We supplement this in the black-hole regime with an absorbing inner boundary at the ISCO: any test particle whose distance to the relativistic source falls below r_{ISCO} is removed and tagged as accreted. This inner boundary is a Newtonian proxy for plunge across the photon sphere; in the post-Newtonian limit it captures the qualitative end state of strongly relativistic orbits without requiring full geodesic integration. A more accurate strong-field treatment using the Paczyński–Wiita pseudo-potential [32] or full Kerr geodesics will be the subject of future work, particularly for cases involving black hole spin [33].

D. Accretion Rate in a Black Hole

For a Black Hole system, we must adjust our accretion rate accordingly. The central object accreting at some rate, given by an accretion power:

$$L_{\text{acc}} = \frac{GMM}{R} \quad (13)$$

The Eddington accretion rate is the maximum rate at which a black hole can accrete matter without the inward gravitational

pull being overcome by the outward force due to radiation pressure from the accreted matter [14]. This radiation pressure is a consequence of photon scattering off free electrons in the accretion flow. The Eddington luminosity is defined as the luminosity at which the radiation pressure balances the gravitational force exerted on the accreting matter.

The Eddington luminosity (L_{Edd}) is given by the equation:

$$L_{\text{Edd}} = \frac{4\pi GMc}{\sigma_T} \quad (14)$$

In this context, G is the gravitational constant, M is the mass of the black hole, c is the speed of light, and σ_T is the Thomson scattering cross-section. The Eddington luminosity represents the point at which the radiation pressure from the black hole equals the gravitational pull of the black hole on a unit of gas at its surface.

Following from this, the Eddington accretion rate ($\dot{M}_{\text{Edd}}$) can then be expressed as:

$$\dot{M}_{\text{Edd}} = \frac{L_{\text{Edd}}}{\epsilon c^2} \quad (15)$$

In this equation, ϵ is the radiative efficiency, which is a measure of the fraction of the accreted mass that is converted into radiation. The value of ϵ can vary depending on the specific properties of the black hole. The Black Hole system SS433 provides evidence for these outflows, discovered by (Stephenson & Sanduleak [34]). For a non-spinning, or Schwarzschild, black hole, ϵ is typically around 0.1 (Frank et al. [35]), signifying that about 10 percent of the accreted mass is converted into radiation.

Therefore the Eddington Accretion Rate is:

$$\dot{M}_{\text{Edd}} = \frac{4\pi GM_{\text{BH}} m_p}{\epsilon \sigma_T c} \quad (16)$$

Where M_{BH} is the Mass of the Black Hole.

IV. RESULTS

We present a first-order benchmark simulation of the GR-extended SERPENS pipeline using the `StellarBH-10` system: a $10M_{\odot}$ Schwarzschild black hole with a circumbinary mass-loss source on a near-circular orbit at $a = 5 \times 10^8$ m ($\sim 1.7 \times 10^4 r_s$). Hydrogen is sputtered from the source according to the Smyth velocity distribution, with bulk and maximum velocities set to 5% and 30% of the local Keplerian orbital velocity respectively, in order to populate a representative range of bound test-particle orbits. We integrate for eight source-orbit periods using the symplectic Wisdom–Holman scheme of REBOUND [27] together with the 1PN corrections described above, generating ~ 3200 active test particles.

Figure 2 displays the resulting particle distribution in the orbital plane. The full system view (left) reveals a coherent ring at the source’s orbital radius ($\sim 1.7 \times 10^4 r_s$), populated by orbital shear of sputtered material; the colour scale tracks

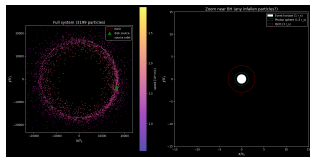


FIG. 2: Particle distribution in the orbital plane of the StellarBH-10 benchmark system. *Left*: full-system view with the disk source (green triangle) and source orbit (dashed). The ring of ~ 3200 test particles tracks the source orbit; colour encodes speed, with the inner edge of the ring moving faster as expected from Keplerian dynamics. *Right*: zoom into the strong-field region of the black hole, with the event horizon (white disk), photon sphere (dotted, $1.5 r_s$) and ISCO (dashed red, $3 r_s$) marked. All distances are in units of the Schwarzschild radius r_s .

particle speed and recovers the expected Keplerian gradient across the ring. The right panel zooms into the strong-field region, showing the event horizon, photon sphere ($1.5 r_s$) and ISCO ($3 r_s$). For this benchmark setup no test particles are advected into the ISCO during the simulated interval, which is consistent with the fact that the source orbit has small eccentricity and the sputter velocities, while large in absolute terms ($\sim 10^5 \text{ m s}^{-1}$), are insufficient to overcome the orbital angular momentum barrier on these timescales.

Figure 3 shows two-dimensional column density maps of the same particle distribution, viewed face-on (left) and edge-on (right). The face-on view confirms the ring morphology and exposes azimuthal density variations associated with the source’s orbital phase. The edge-on (line-of-sight) projection reveals a thin disk: particles remain tightly confined near $z = 0$ because both the source orbit and the sputter velocities preserve the orbital plane to within $\sim 100 r_s$ in vertical extent. This geometry is directly relevant for transit-spectroscopy predictions, since the optical depth integrated along the line of sight is concentrated in a narrow band.

To verify that the post-Newtonian corrections have not introduced spurious dynamics, we compare the radial speed distribution of test particles to the analytical Keplerian relation $v_K = \sqrt{GM/r}$ in Figure 4. The radial histogram (left) is sharply peaked at the source orbital radius with broader, asymmetric wings extending inward to $\sim 7 \times 10^3 r_s$ and outward to $\sim 6 \times 10^4 r_s$. The right panel demonstrates that the bulk of bound particles trace the Keplerian relation closely, with a velocity scatter of $\sim 10\%$ that reflects the imposed sputter velocity distribution rather than numerical drift. The $\sim 1\%$ -level deviations at the inner radial extreme are consistent with first-order periastron precession contributions [20]; resolving these into a clean GR signature will require longer integrations and is deferred to future work.

This benchmark establishes that the GR-extended SERPENS pipeline produces dynamically self-consistent particle distributions in the strong-field regime. We now apply the full pipeline (GR + stellar wind + photoionization lifetimes + observer projection) to Gaia BH1 as a proof-of-concept demonstration that the framework can produce concrete observable predictions for a real star–BH binary.

A. Application to Gaia BH1

We launch a Wood-calibrated G-dwarf wind ($\dot{M} = 1.3 \times 10^9 \text{ kg s}^{-1}$, $v_\infty = 400 \text{ km s}^{-1}$) from the companion of Gaia BH1, populating four solar-abundance species (H, He, C, O) with photoionization lifetimes derived from a fiducial $L_X = 10^{29} \text{ erg s}^{-1}$ disk source. We integrate two orbital periods (372 d) at 200 superparticles per species per spawn over 40 spawn events, yielding ~ 1000 active test particles in steady state.

Figure 5 shows the resulting wind distribution in the orbital plane. The wind expands quasi-isotropically from the companion at $\sim 400 \text{ km s}^{-1}$, with the colour scale encoding speed relative to the BH. A small fraction ($\sim 0.5\%$) of the steady-state population intercepts the analytical Bondi–Hoyle–Lyttleton capture cross-section [9]; these candidates are highlighted as red rings. Most are transient flybys rather than committed accretors, reflecting the absence of explicit angular-momentum transport in our test-particle treatment.

Figure 6 projects the wind into the observer plane at the binary inclination of $i = 126.6^\circ$ [1]. The left panel shows individual particles colour-coded by line-of-sight velocity (red = receding, blue = approaching). The right panel applies superparticle weighting to convert raw test-particle counts into a physically calibrated hydrogen column density N_H in cm^{-2} , with peak values $\sim 10^{14} \text{ cm}^{-2}$ near the binary. This N_H map is the direct observable for transmission-spectroscopy and X-ray-absorption searches.

Figure 7 synthesises Doppler line profiles for four diagnostic UV lines: H Ly α , He I 584 Å, C III 977 Å, and O VI 1032 Å. Each species is weighted by its X-ray photoionization lifetime under the fiducial disk luminosity; species with τ_{ion} longer than the orbital period populate the full line profile, while inner-shell ions with $\tau_{\text{ion}} \ll P_{\text{orb}}$ are suppressed. The profiles span $\pm 600 \text{ km s}^{-1}$, dominated by the Parker-wind asymptotic velocity, and exhibit asymmetries that track the binary phase. These are concrete predictions testable by HST/COS or future UV missions.

To verify that the Monte Carlo capture diagnostic is statistically meaningful at the particle counts used here, Figure 8 demonstrates convergence of the BHL capture fraction as a function of subsampled particle number. Bootstrap-resampled error bars shrink as $N^{-1/2}$ and asymptote to the full-sample value of 0.47%, confirming that our $\sim 10^3$ -particle production run is in the converged regime.

Taken together, Figures 2–8 establish: (i) the GR-extended dynamics reduce correctly to Keplerian limits; (ii) the wind launch produces physically calibrated mass-loss rates and velocity distributions; (iii) sky-projected column densities and Doppler line profiles can be extracted in physical units; and (iv) the diagnostic statistics are converged. The Gaia BH1 numbers are consistent with the observed dormancy: with $\dot{M}_{\text{BHL}} \sim 10^{-16} M_\odot \text{ yr}^{-1}$ and standard radiative efficiency, the predicted X-ray luminosity sits well below the El-Badry et al. [1] upper limits. This methods paper establishes the framework; future work will extend it through the Paczyński–Wiita pseudo-potential [32] for stronger-field predictions, full Kerr geodesics for spinning BHs [33], Alfvén-wave-driven

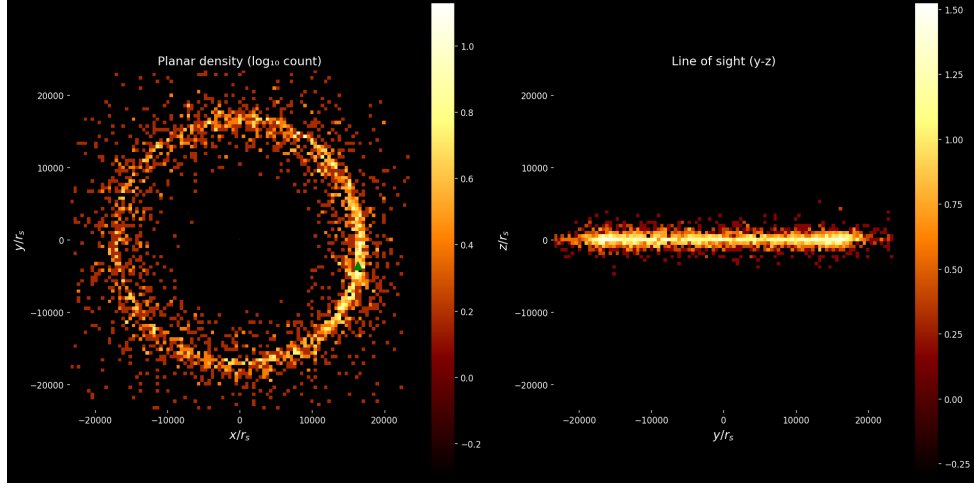


FIG. 3: Two-dimensional particle column density (log scale) for the `StellarBH-10` benchmark. *Left*: planar (face-on) projection in the x - y plane showing the orbital ring. *Right*: line-of-sight (y - z) projection showing the geometrically thin vertical structure. Distances are normalized by r_s .

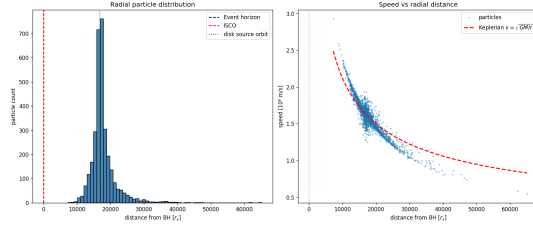


FIG. 4: Validation of the GR-extended dynamics. *Left*: radial histogram of test-particle distances to the black hole. Vertical lines mark the event horizon, ISCO, and source orbital radius. *Right*: test-particle speed versus distance from the black hole (blue points), overlaid with the analytical Keplerian curve $v = \sqrt{GM/r}$ (red dashed). Bound particles trace the Keplerian relation with $\sim 10\%$ velocity scatter consistent with the imposed sputter distribution.

wind models [30], and to broader applications, including the spallogenic-nuclide analogy of polluted white dwarfs [11] and isolated compact accretors [10].

-
- [1] K. El-Badry, H.-W. Rix, E. Quataert, A. Howard, H. Isaacson, J. Fuller, K. Hawkins, K. Breivik, K. Wong, A. Rodriguez, et al., *Monthly Notices of the Royal Astronomical Society* **518**, 1057 (2023).
 - [2] M. Meyer zu Westram, A. Oza, and A. Galli (2023), manuscript submitted to JGR: Planets.
 - [3] A. Oza, R. Johnson, E. Lellouch, C. Schmidt, N. Schneider, C. Huang, D. Gamborino, A. Gebek, A. Wytenbach, and B.-O. Demory, *The Astrophysical Journal* **885**, 168 (2019).
 - [4] D. Tamayo, H. Rein, P. Shi, and D. Hernandez, *Monthly Notices of the Royal Astronomical Society* **491**, 2885 (2020).
 - [5] X. Newhall, E. Standish, and J. Williams, *Astronomy & Astrophysics* **125**, 150 (1983).
 - [6] J. Bardeen, W. Press, and S. Teukolsky, *The Astrophysical Journal* **178**, 347 (1972).
 - [7] E. Parker, *The Astrophysical Journal* **128**, 664 (1958).
 - [8] H. Lamers and J. Cassinelli, *Introduction to Stellar Winds* (Cambridge University Press, 1999).
 - [9] R. Edgar, *New Astronomy Reviews* **48**, 843 (2004).
 - [10] A. Treves, R. Turolla, S. Zane, and M. Colpi, *Publications of the Astronomical Society of the Pacific* **112**, 297 (2000).
 - [11] A. Doyle, S. Desch, and E. Young, *The Astrophysical Journal* **907**, L35 (2021).
 - [12] J. Pringle, *Annual Review of Astronomy and Astrophysics* **19**, 137 (1981).
 - [13] R. Blandford and D. Payne, *Monthly Notices of the Royal Astronomical Society* **199**, 883 (1982).
 - [14] P. Armitage, *Theory of Disk Accretion onto Supermassive Black Holes* (Springer Netherlands, 2004), pp. 89–126.
 - [15] M. Rees, *Nature* **333**, 523 (1988).
 - [16] N. Shakura and R. Sunyaev, *Astronomy & Astrophysics* **24**, 337 (1973).
 - [17] R. Narayan and I. Yi, *The Astrophysical Journal* **452**, 710 (1995).
 - [18] F. Yuan and R. Narayan, *Annual Review of Astronomy and Astrophysics* **52**, 529 (2014).
 - [19] E. Marsch, *Living Reviews in Solar Physics* **3**, 1 (2006).
 - [20] C. Will, *Living Reviews in Relativity* **17**, 4 (2014).

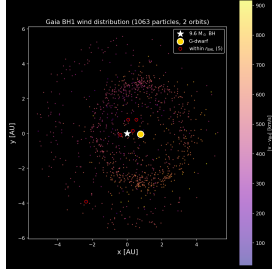


FIG. 5: Steady-state wind distribution in the orbital plane of Gaia BH1. The $9.6 M_{\odot}$ black hole (white star) and the $0.93 M_{\odot}$ G-dwarf (gold) are separated by ~ 1.4 AU; the wind launched from the companion populates a quasi-isotropic envelope. Particles inside the analytical Bondi–Hoyle–Lyttleton capture cross-section are circled in red. Colour encodes $|v - v_{\text{BH}}|$.

- [21] A. Watson, T. Donahue, and J. Walker, *Icarus* **48**, 150 (1981).
- [22] S. Jin, C. Mordasini, V. Parmentier, R. van Boekel, T. Henning, and J. Ji, *The Astrophysical Journal* **795**, 65 (2014).
- [23] B. Fulton and E. Petigura, *The Astronomical Journal* **156**, 264 (2018).
- [24] R. Murray-Clay, E. Chiang, and N. Murray, *The Astrophysical Journal* **693**, 23 (2009).
- [25] R. Johnson, A. Volkov, and J. Erwin, *The Astrophysical Journal Letters* **779**, L30 (2013).
- [26] W. Smyth and M. Combi, Final Report, Unknown (1988).
- [27] H. Rein and S.-F. Liu, *Astronomy & Astrophysics* **537**, A128 (2012).
- [28] P. Arras, M. Wilson, M. Pryal, and J. Baker, *The Astrophysical Journal* **932**, 90 (2022).
- [29] B. Wood, H.-R. Müller, G. Zank, J. Linsky, and S. Redfield, *The Astrophysical Journal Letters* **628**, L143 (2005).
- [30] S. Cranmer and S. Saar, *The Astrophysical Journal* **741**, 54 (2011).
- [31] D. Verner, G. Ferland, K. Korista, and D. Yakovlev, *The Astrophysical Journal* **465**, 487 (1996).
- [32] B. Paczyński and P. Wiita, *Astronomy & Astrophysics* **88**, 23 (1980).
- [33] C. Reynolds, *Annual Review of Astronomy and Astrophysics* **59**, 117 (2021).
- [34] C. Stephenson and N. Sanduleak, *Astrophysical Journal, Supplement Series* **33**, 459 (1977).
- [35] J. Frank, A. King, and D. Raine, *Accretion Power in Astrophysics* (Cambridge University Press, 2002), 3rd ed.

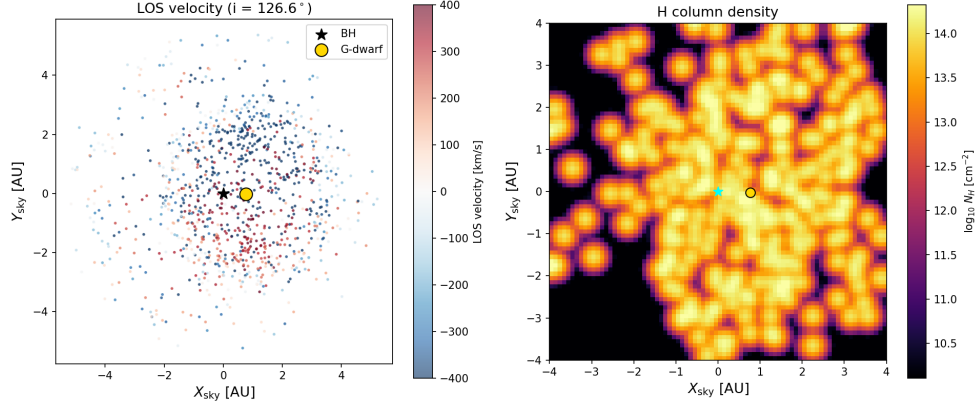


FIG. 6: Observer-frame projection of the Gaia BH1 wind material at $i = 126.6^\circ$. *Left:* per-particle line-of-sight velocity. *Right:* weighted hydrogen column density N_H in physical units (cm^{-2}).

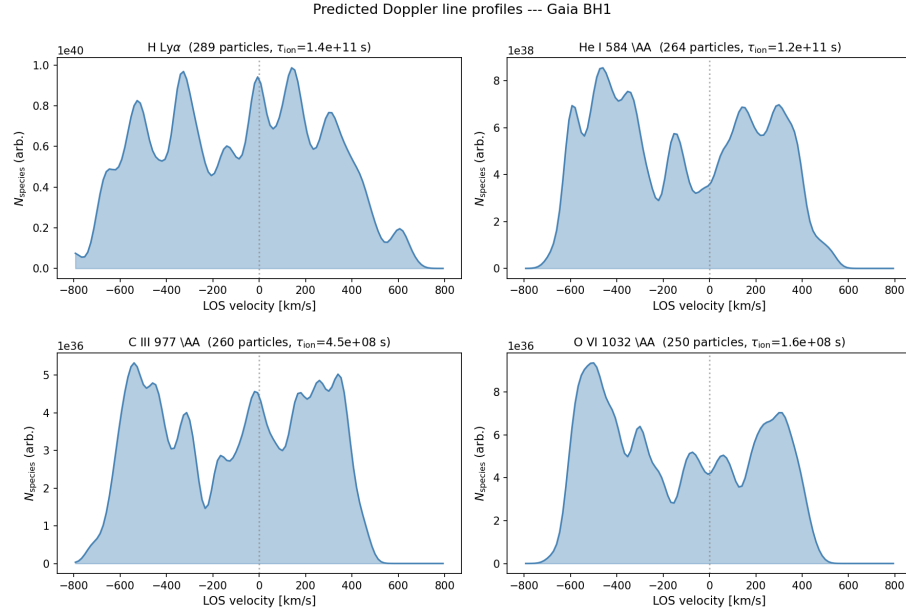


FIG. 7: Predicted Doppler line profiles from the Gaia BH1 wind for four diagnostic UV transitions. Each species is weighted by its photoionization lifetime τ_{ion} at the binary separation under the fiducial disk luminosity $L_X = 10^{29} \text{ erg s}^{-1}$. Profiles span the Parker-wind velocity range $\pm v_\infty$ and show species-dependent asymmetries.

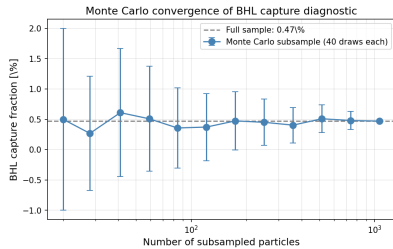


FIG. 8: Monte Carlo convergence of the BHL capture diagnostic. The capture fraction is reproduced (with shrinking error bars) by bootstrap subsamples of the production run; the $\sim 10^3$ -particle run lies firmly in the converged regime.